**Name:** ______________________ **Date:** ______________________

**Collaborators:** ______________________ **Lab Section:** ______________

## Postlab Exercises (20 points)

*Submit the postlab to the TA at the end of the lab.*

### Ohm's Law (15 points)

**Question 3** **½ point**

Enter the single-resistor data into Data Table 1.

Data table 1: *Single resistor data.*

$R =$ ______________

| **Power Supply Setting** | **Measured Voltage** (V) | **Measured Current** (A) |
|---|---|---|
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**Question 4** **$^1/_2$ point**

Enter the light bulb data into Data Table 2.

Data table 2: *Bulb data.*

| **Power Supply Setting** | **Measured Voltage** (V) | **Measured Current** (A) |
|---|---|---|
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**Question 5** **$^1/_2$ point**

Enter the series resistor data into Data Table 3.

Data table 3: *Series resistor data.*

| | |
|---|---|
| $R_1$ | |
| $R_2$ | |
| $V_s$ | |
| $I_s$ | |
| $V_1$ | |
| $V_2$ | |

**Question 6** **$^1/_2$ point**

Enter the parallel resistor data into Data Table 4.

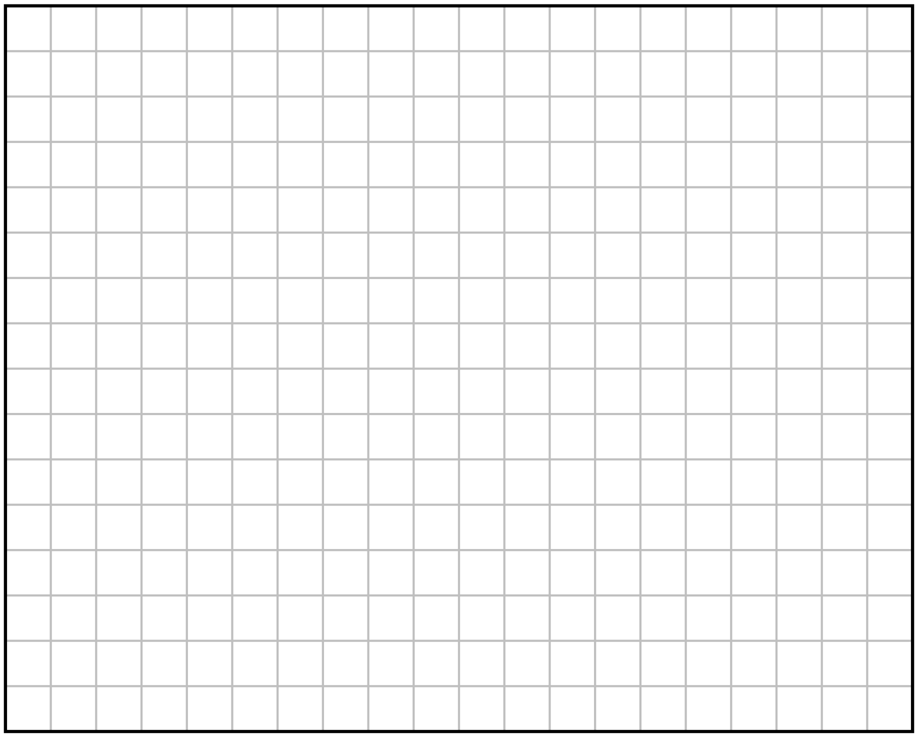
Data table 4: *Parallel resistor data.*

| | |
|---|---|
| $R_1$ | |
| $R_2$ | |
| $V_s$ | |
| $I_s$ | |
| $I_1$ | |
| $I_2$ | |

**Question 7** **2 points**

**Single resistor analysis**: take the single resistor data from Data Table 1 and plot voltage versus current in Graph 1, including labels and units. Is the relationship linear? If so, draw the best-fit line.

Graph 1: $V$ *vs.* $I$ *for the single resistor.*



**Question 8** **1 point**

Calculate the slope of the best-fit line in Graph 1 and use it to estimate the value of $R$. Compare this to the value labeled on the actual resistor and the value you measured from the multimeter.

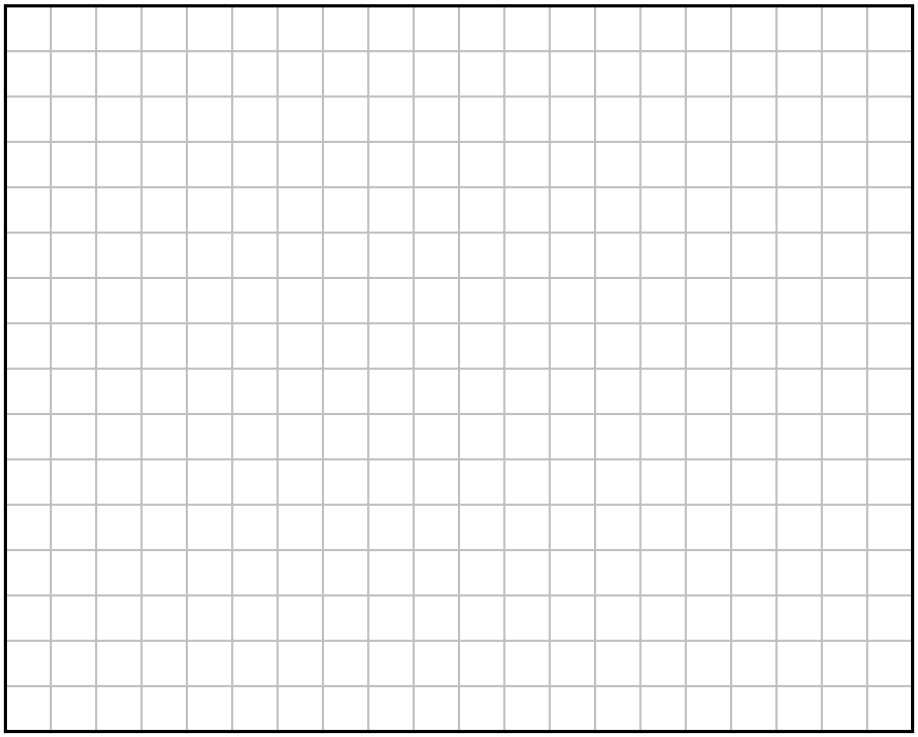
**Question 9** **1 point**

Calculate the power dissipated in the resistor when $V = 6.0\,\mathrm{V}$.

**Question 10** **2 points**

**Single bulb analysis**: take the single resistor data from Data Table 2 and plot voltage versus current in Graph 2, including labels and units.

Graph 2: *$V$ vs. $I$ for the light bulb.*



**Question 11** **1 point**

Is there a linear relationship between voltage and current in Graph 2, compared to the linearity of the single resistor in Graph 1? (It should not be!)

**Question 12** **1 point**

Calculate the resistance of the filament when $V = 6.0\,\text{V}$ and when $V = 1.0\,\text{V}$. Note that a non-linear function does not have a constant slope! To calculate the resistance at a certain voltage, draw a line tangent to the curve at that voltage. Then calculate the slope of the tangent line to find the instantaneous resistance.

**Question 13** **1 point**

Intuitively, do you think the temperature of the filament is higher at $V = 6.0\,\text{V}$ and $V = 1.0\,\text{V}$? How does this information compare with what you learned about metal superconductors as the temperature decreases?

**Question 14** **1 point**

**Series resistance analysis**: Use eq. (4) and verify if your measured $V_1$ is comparable to the expected value. Show all your work.

**Question 15** **1 point**

From the value of $I_s$ and $V_s$, calculate the equivalent or total resistance of the series resistor circuit. Compare this with the theoretical value using eq. (2).

**Question 16** **1 point**

**Parallel resistance analysis**: use eq. (5) and verify if your measured $I_1$ is comparable to the expected value. Show all your work.

**Question 17** **1 point**

From the value of $I_s$ and $V_s$, calculate the equivalent or total resistance of the parallel resistor circuit. Compare this with the theoretical value using eq. (3).

## Superconductivity (5 points)

**Question 18** **2 points**

Fill in the measurements of the superconducting transition made as the superconductor is allowed to warm up. Use Table 2 to convert the measured thermal conductivity $\kappa$ in mV to superconductor temperature $T$ in K. Use Ohm's law to convert voltage and current measurements to resistance.

Data table 5: *Superconductor Temperature Data.*

| **Thermal Conductivity** (mV) | **Temperature** (K) | **Voltage** (V) | **Current** (A) | **Resistance** ($\Omega$) |
|---|---|---|---|---|
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Table 2: *Conversion of thermal conductivity $\kappa$ to superconductor temperature $T$.*

| $\kappa$ (mV) | $T$ (K) | $\kappa$ (mV) | $T$ (K) | $\kappa$ (mV) | $T$ (K) | $\kappa$ (mV) | $T$ (K) | $\kappa$ (mV) | $T$ (K) | $\kappa$ (mV) | $T$ (K) | $\kappa$ (mV) | $T$ (K) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6.92 | 70 | 6.29 | 80 | 5.9 | 90 | 5.52 | 100 | 5.16 | 110 | 4.81 | 120 | 4.46 | 130 |
| 6.85 | 71 | 6.25 | 81 | 5.86 | 91 | 5.48 | 101 | 5.13 | 111 | 4.77 | 121 | 4.42 | 131 |
| 6.78 | 72 | 6.21 | 82 | 5.83 | 92 | 5.44 | 102 | 5.09 | 112 | 4.74 | 122 | 4.39 | 132 |
| 6.71 | 73 | 6.17 | 83 | 5.79 | 93 | 5.41 | 103 | 5.06 | 113 | 4.70 | 123 | 4.35 | 133 |
| 6.64 | 74 | 6.13 | 84 | 5.75 | 94 | 5.37 | 104 | 5.02 | 114 | 4.67 | 124 | 4.32 | 134 |
| 6.56 | 75 | 6.09 | 85 | 5.72 | 95 | 5.34 | 105 | 4.99 | 115 | 4.63 | 125 | 4.28 | 135 |
| 6.49 | 76 | 6.05 | 86 | 5.68 | 96 | 5.30 | 106 | 4.95 | 116 | 4.60 | 126 | 4.25 | 136 |
| 6.42 | 77 | 6.01 | 87 | 5.64 | 97 | 5.27 | 107 | 4.91 | 117 | 4.56 | 127 | 4.21 | 137 |
| 6.37 | 78 | 5.97 | 88 | 5.6 | 98 | 5.23 | 108 | 4.88 | 118 | 4.53 | 128 | 4.18 | 138 |
| 6.33 | 79 | 5.93 | 89 | 5.56 | 99 | 5.20 | 109 | 4.84 | 119 | 4.49 | 129 | 4.14 | 139 |

**Question 19** **3 points**

Take your measurements from Data Table 5 and plot of resistance versus temperature in Graph 3. Include labels and units. Do not draw a line of best fit. Mark on your plot the temperature where the levitating magnet fell. What is the critical temperature $T_C$ of the superconductor?

Graph 3: *$R$ vs. $T$ for the Superconductor.*